UNIVERSITÀ DI TRENTO

The Devil Is in Gradient Entanglement: Energy-Aware Gradient Coordinator for Robust Generalized Category Discovery

CVPR JUNE 3-7, 2026 DENVER COLORADO

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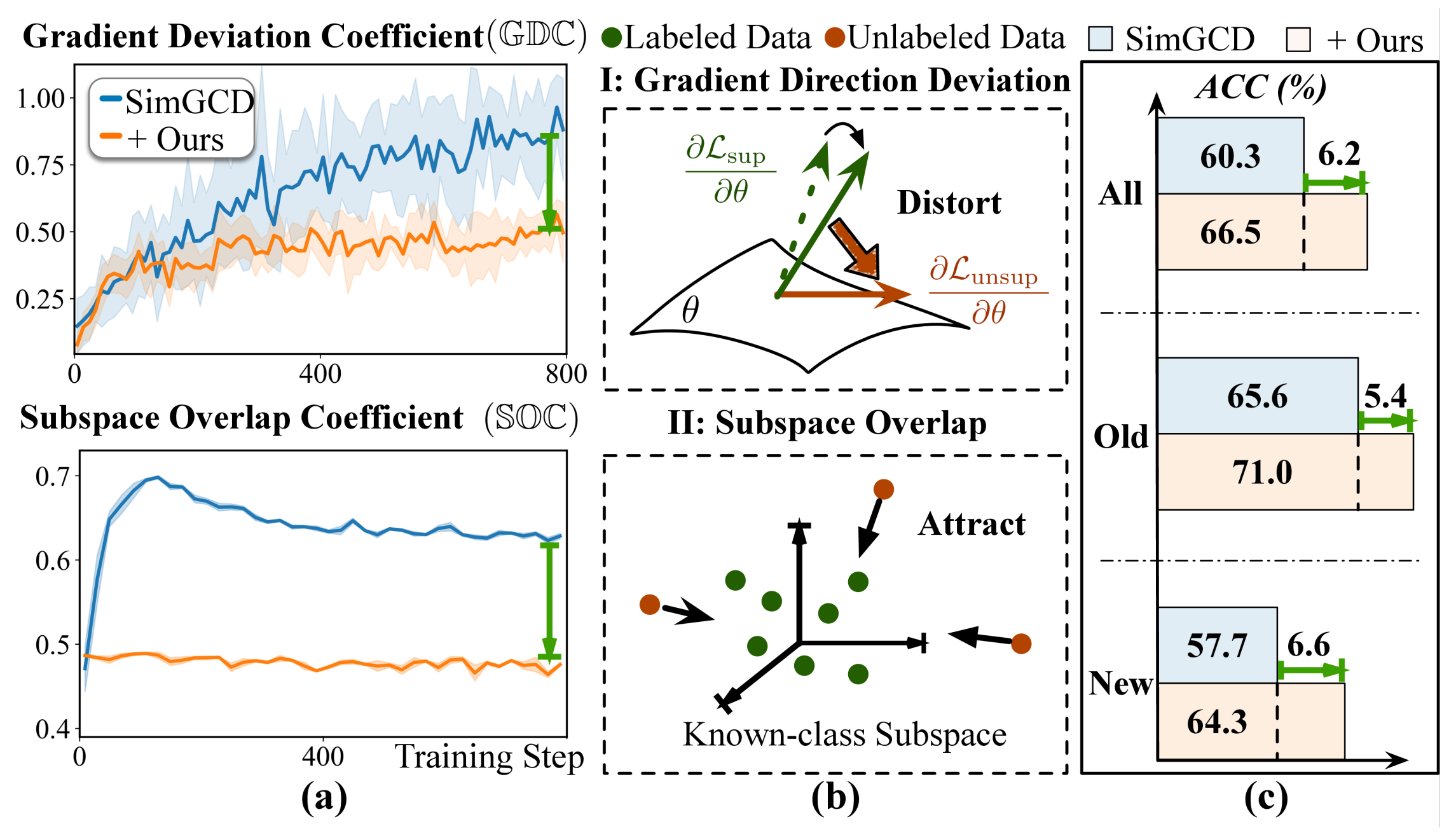
1. Problem: Gradient Entanglement

Observation. GCD jointly optimizes labeled and unlabeled objectives:

$$\mathcal{L}_{\text{GCD}} = \alpha\mathcal{L}_{\text{sup}}(\mathcal{D}_L) + \beta\mathcal{L}_{\text{unsup}}(\mathcal{D}_U), \qquad g = g_L + g_U.$$

Gradient Entanglement (GE): unlabeled gradients distort the supervised direction and attract novel features toward the known-class subspace.



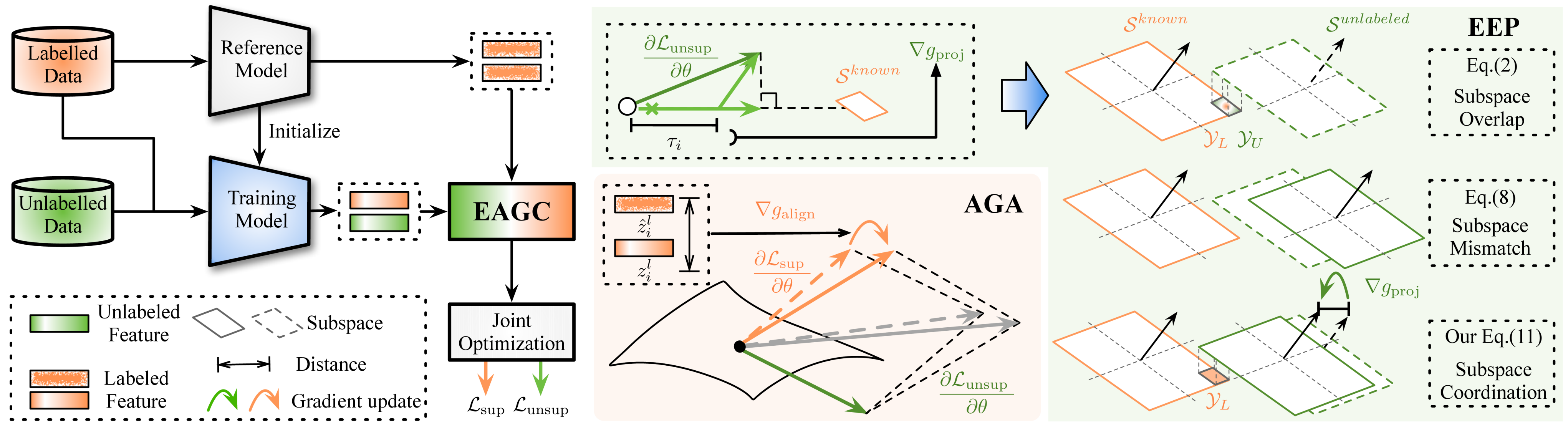
Direction deviation

unreliable update direction

Subspace overlap

novel features overlap known subspace

2. Method: Energy-Aware Gradient Coordinator


Labelled Data
Reference Model
Initialize
Unlabelled Data
Training Model
EAGC
Joint Optimization
$\mathcal{L}_{\text{sup}}$ $\mathcal{L}_{\text{unsup}}$
Unlabeled Feature
Labeled Feature
Subspace
Distance
Gradient update
$\frac{\partial \mathcal{L}_{\text{unsup}}}{\partial \theta}$
τ_i
$\mathcal{S}^{known}$
∇g_{proj}
AGA
$\hat{z}_i^l$
z_i^l
∇g_{align}
$\frac{\partial \mathcal{L}_{\text{sup}}}{\partial \theta}$
$\frac{\partial \mathcal{L}_{\text{unsup}}}{\partial \theta}$
$\mathcal{S}^{known}$
$\mathcal{S}^{unlabeled}$
$\mathcal{Y}_L$ $\mathcal{Y}_U$
$\mathcal{Y}_L$
EEP
Eq.(2) Subspace Overlap
Eq.(8) Subspace Mismatch
∇g_{proj}
Our Eq.(11) Subspace Coordination

Anchor-based Gradient Alignment (AGA)

Train a reference encoder on $\mathcal{D}_L$ only, freeze it, then anchor labeled gradients to stable supervised features:

$$\nabla_{z^l} g_{\text{align}} = \lambda_a(z^l - \hat{z}^l).$$

Energy-aware Elastic Projection (EEP)

Build a soft known-class subspace and project unlabeled gradients with sample-adaptive strength:

$$\nabla_{z^u} g_{\text{proj}} = -\lambda_p(\tau \cdot \nabla_{z^u}\mathcal{L}_{\text{unsup}}\mathcal{S}^{\text{known}}).$$

AGA stabilizes known classes; EEP decouples likely novel samples without over-suppressing likely known samples.

3. Quantitative Comparison

Main SOTA comparison. We plug EAGC into four representative baselines across five datasets, showing broad improvements in All/New ACC without changing the inference model.

Method	CUB			Stanford-Cars			Aircraft			CIFAR-100			ImageNet-100			Average		
	All	Old	New	All	Old	New	All	Old	New	All	Old	New	All	Old	New	All	Old	New
ORCA [2]	36.3	43.8	32.6	31.6	32.0	31.4	31.9	42.2	26.9	73.5	**92.6**	63.9	81.8	86.2	79.6	51.0	59.4	46.9
GCD [63]	51.3	56.6	48.7	39.0	57.6	29.9	45.0	41.1	46.9	73.0	76.2	66.5	74.1	89.8	66.3	56.5	64.3	51.7
GPC [88]	52.0	55.5	47.5	38.2	58.9	27.4	43.3	40.7	44.8	75.4	84.6	60.1	75.3	93.4	66.7	56.8	66.6	49.3
XCon [10]	52.1	54.3	51.0	40.5	58.8	31.7	47.7	44.4	49.4	74.2	81.2	60.3	77.6	93.5	69.7	58.4	66.4	52.4
PromptCAL [85]	62.9	64.4	62.1	50.2	70.1	40.6	52.2	52.2	52.3	81.2	84.2	75.3	83.1	92.7	78.3	65.9	72.7	61.7
DCCL [43]	63.5	60.8	64.9	43.1	55.7	36.2	–	–	–	75.3	76.8	70.2	80.5	90.5	76.2	–	–	–
μGCD [65]	65.7	68.0	64.6	56.5	68.1	50.9	53.8	55.4	53.0	–	–	–	–	–	–	–	–	–
InfoSieve [49]	69.4	**77.9**	65.2	55.7	74.8	46.4	56.3	63.7	52.5	78.3	82.2	70.5	80.5	93.8	73.8	68.0	78.5	61.7
AMEND [1]	64.9	75.6	59.6	56.4	73.3	48.2	52.8	61.8	48.3	81.0	79.9	83.3	83.2	92.9	78.3	67.7	76.7	63.5
CMS [6]	68.2	76.5	64.0	56.9	76.1	47.6	56.0	63.4	52.3	82.3	85.7	75.5	84.7	95.6	79.2	69.6	79.5	63.7
SimGCD [74]	60.3	65.6	57.7	53.8	71.9	45.0	54.2	59.1	51.8	80.1	81.2	77.8	83.0	93.1	77.9	66.3	74.2	62.0
+ EAGC	66.5	71.0	64.3	62.9	76.0	56.6	57.7	60.4	56.3	83.1	84.1	81.0	83.5	93.9	78.3	70.7	77.1	67.3
LegoGCD [3]	63.8	71.9	59.8	57.3	75.7	48.4	55.0	61.5	51.7	81.8	81.4	82.5	86.3	94.5	82.1	68.8	77.0	64.9
+ EAGC	64.6	72.5	60.7	62.8	76.2	56.3	56.7	61.9	54.0	83.5	83.7	83.0	84.8	94.4	80.1	70.5	77.7	66.8
SPTNet [68]	65.8	68.8	65.1	59.0	79.2	49.3	59.3	61.8	58.1	81.3	84.3	75.6	85.4	93.2	81.4	70.2	77.5	65.9
+ EAGC	67.4	70.5	66.0	62.1	76.1	55.4	57.0	58.1	56.5	**84.0**	84.3	**83.5**	85.3	94.3	80.4	71.2	76.7	68.4
SelEx [50]	73.6	75.3	72.8	58.5	75.6	50.3	57.1	64.7	53.3	82.3	85.3	76.3	83.1	93.6	77.8	70.9	78.9	66.1
+ EAGC	**83.2**	73.9	**87.9**	**65.7**	**83.7**	**57.0**	**65.7**	**65.6**	**65.7**	79.3	84.1	69.7	**89.8**	**96.2**	**86.5**	**76.7**	**80.7**	**73.4**
Avg. △	+4.6	+1.6	+5.9	+6.2	+2.4	+8.1	+2.9	-0.3	+4.4	+1.1	+1.0	+1.3	+1.4	+1.1	+1.5	+3.2	+1.2	+4.3

4. Diagnostics and Robustness

Gradient Entanglement and backbone robustness. GE is measured by direction deviation $\mathbb{GDC}$ and subspace overlap $\mathbb{SOC}$; DINOv2 tests whether the gain transfers across backbones.

GE metrics

Method	CUB		Stanford Cars		Aircraft	
	$\mathbb{GDC}\downarrow$	$\mathbb{SOC}\downarrow$	$\mathbb{GDC}\downarrow$	$\mathbb{SOC}\downarrow$	$\mathbb{GDC}\downarrow$	$\mathbb{SOC}\downarrow$
SimGCD [74]	0.2669	0.6478	0.0909	0.8110	0.0935	0.8643
+EAGC	**0.1525**	**0.4923**	**0.0500**	**0.6208**	**0.0375**	**0.7139**
LegoGCD [3]	0.2646	0.6468	0.0905	0.8112	0.0922	0.8642
+EAGC	**0.1594**	**0.4844**	**0.0515**	**0.6135**	**0.0376**	**0.7112**
SelEx [50]	0.0005	0.5902	0.0007	0.6022	0.0313	0.6033
+EAGC	**0.0003**	**0.4463**	**0.0001**	**0.4470**	**0.0001**	**0.5774**

GE metrics. Lower $\mathbb{GDC}$ and $\mathbb{SOC}$ show that EAGC reduces direction deviation and known-novel subspace overlap on all three fine-grained datasets.

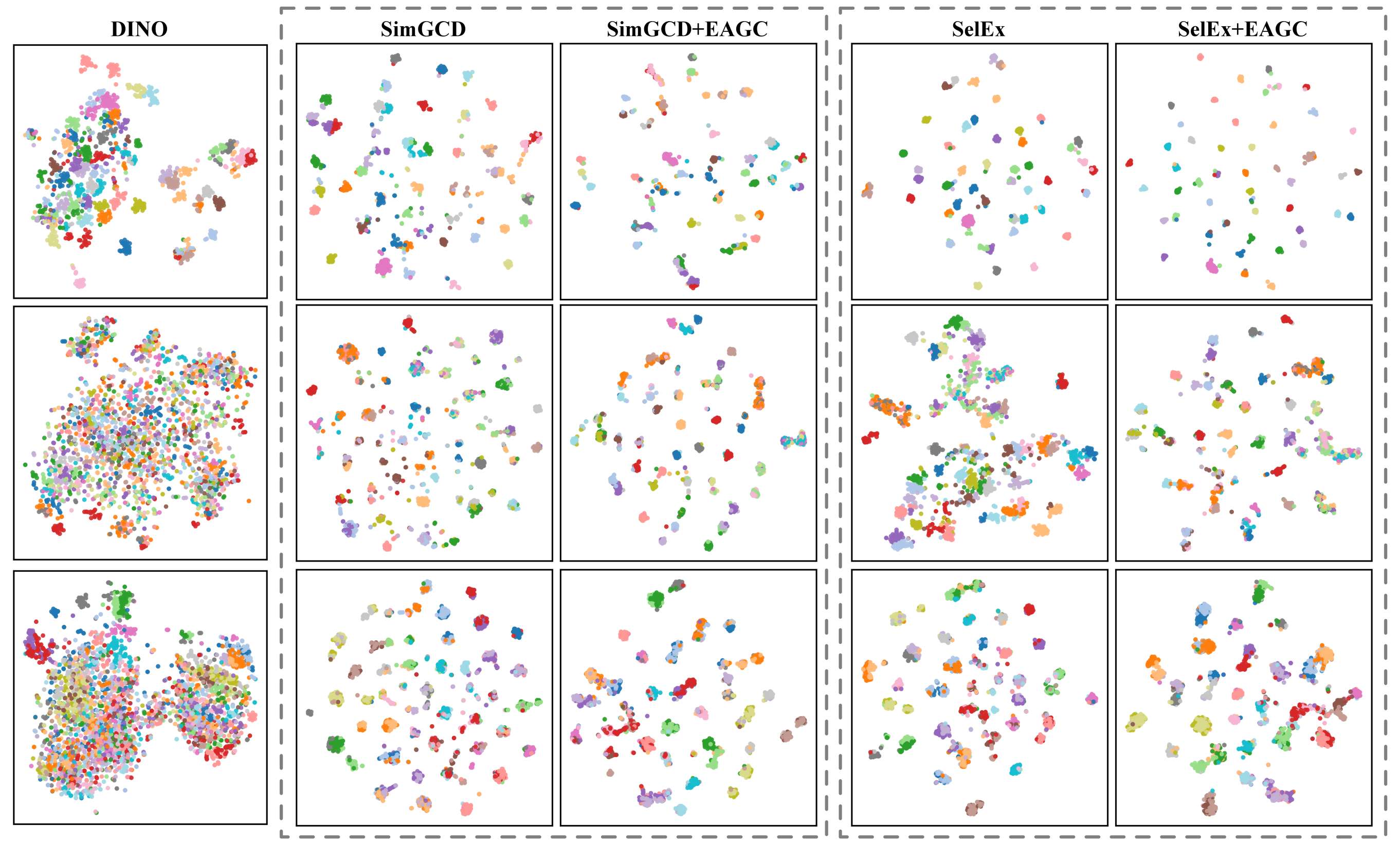
DINOv2 results

Method	CUB			Stanford Cars			Aircraft		
	All	Old	New	All	Old	New	All	Old	New
SimGCD	71.5	78.1	68.3	71.5	81.9	66.6	63.9	69.9	60.9
+ EAGC	**77.9**	**79.9**	**76.9**	**81.5**	**90.9**	**76.9**	**74.6**	**74.9**	**74.4**
SelEx	87.4	**85.1**	88.5	82.2	**93.7**	76.7	79.8	**82.3**	78.6
+ EAGC	**89.4**	82.3	**92.9**	**85.3**	92.7	**81.7**	**84.9**	81.8	**86.5**
Avg. Δ	+4.2	-0.5	+6.5	+6.6	+4.0	+7.7	+7.9	+2.3	+10.7

Backbone robustness. With DINOv2 features, the same training-time coordinator still improves average All/New ACC by $+6.2/+8.3$ over the baseline.

5. Representation Visualization

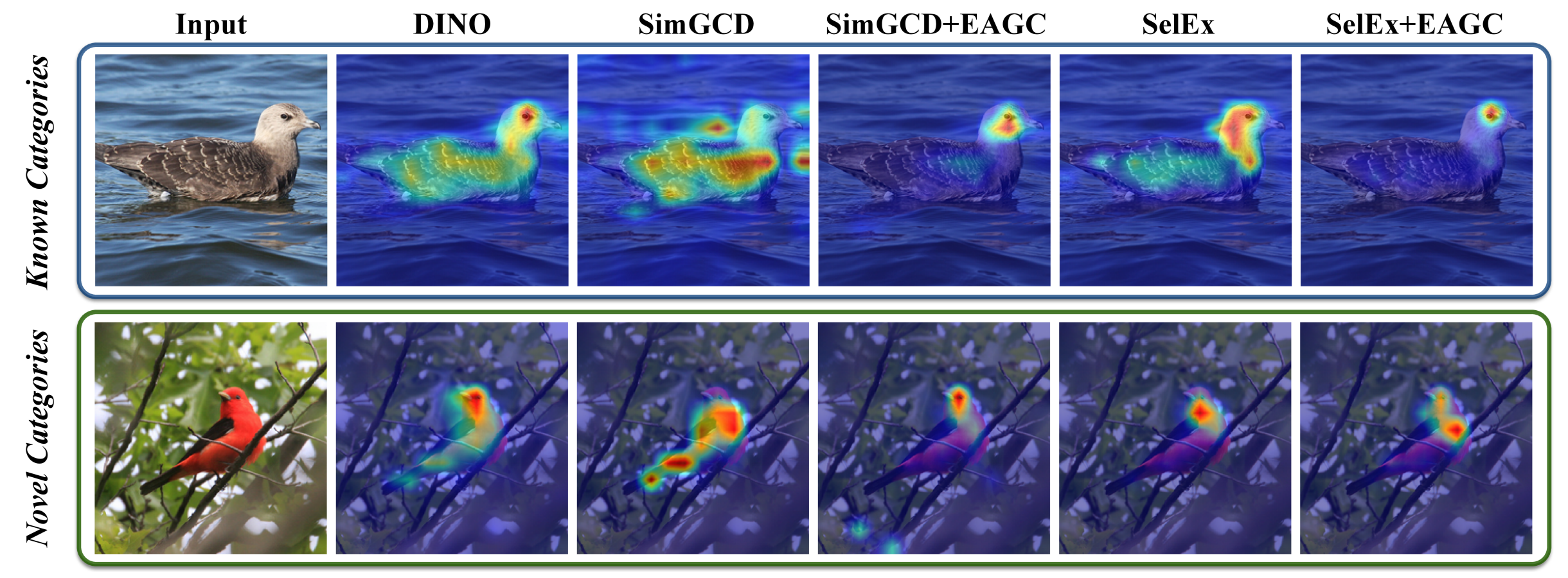
t-SNE visualization. EAGC produces cleaner and better separated feature clusters, consistent with the reduced subspace overlap measured above.



- Better known-novel separation.
- Cleaner clusters across datasets.
- Consistent with reduced GE.

6. Qualitative Evidence and Project

Visual evidence. Attention maps show how EAGC-guided learning focuses on more discriminative object regions.



Plug-and-play training

- No network architecture change.
- No base-objective rewrite.
- Training-time only; zero inference cost.

haiyangzheng.github.io/EAGC

Takeaway: coordinate gradients to preserve known-class discrimination and improve novel-class discovery.